

```
import os
os.listdir()

['.config', 'Advertising.csv', 'sample_data']
```

```
import pandas as pd

# Load Dataset
df = pd.read_csv('Advertising.csv')

# Show first 5 rows
df.head()
```

	Unnamed: 0	TV	Radio	Newspaper	Sales
0	1	230.1	37.8	69.2	22.1
1	2	44.5	39.3	45.1	10.4
2	3	17.2	45.9	69.3	9.3
3	4	151.5	41.3	58.5	18.5
4	5	180.8	10.8	58.4	12.9

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
# Dataset Information
df.info()

# Statistical Summary
df.describe()

# Check Missing Values
df.isnull().sum()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 5 columns):
#   Column      Non-Null Count  Dtype
---  -
0   Unnamed: 0   200 non-null   int64
1   TV           200 non-null   float64
2   Radio        200 non-null   float64
3   Newspaper    200 non-null   float64
4   Sales        200 non-null   float64
dtypes: float64(4), int64(1)
memory usage: 7.9 KB

0

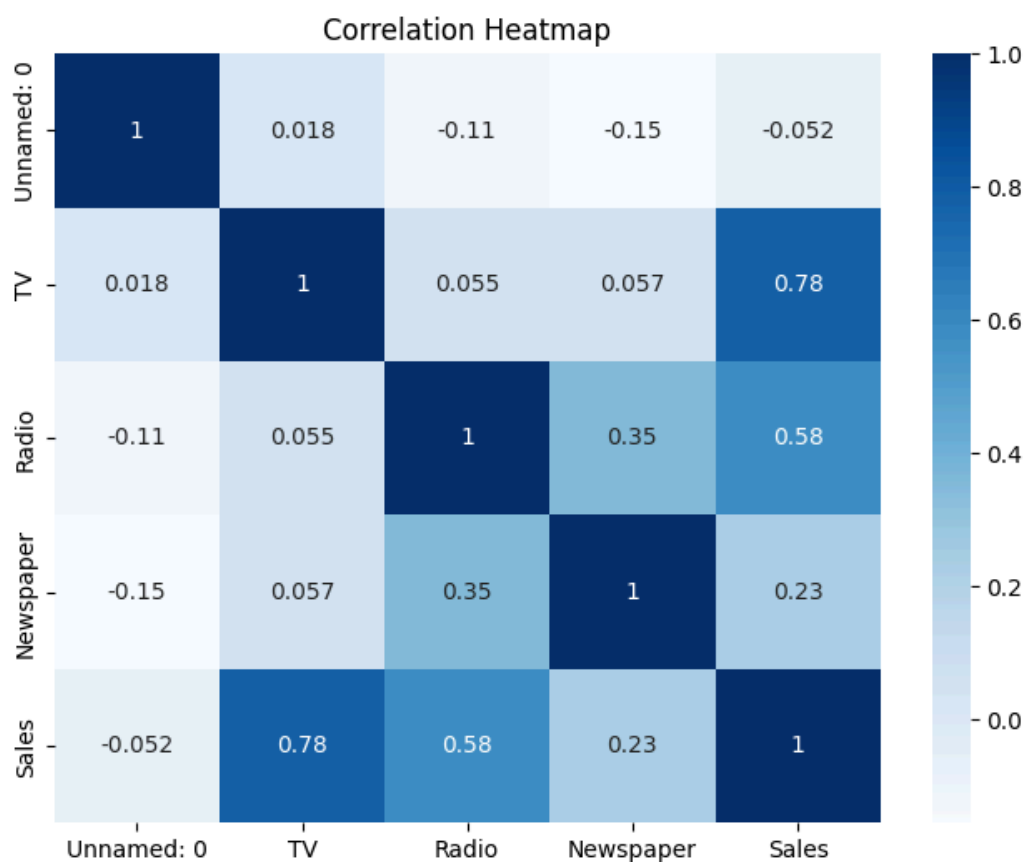
```

Unnamed: 0	0
TV	0
Radio	0
Newspaper	0
Sales	0

```
dtype: int64
```

```
import matplotlib.pyplot as plt
import seaborn as sns

# Correlation Heatmap
plt.figure(figsize=(8,6))
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap='Blues')
plt.title("Correlation Heatmap")
plt.show()
```



```
from sklearn.model_selection import train_test_split

# Features and Target
X = df[['TV', 'Radio', 'Newspaper']]
y = df['Sales']

# Train-Test Split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

```
from sklearn.linear_model import LinearRegression

model = LinearRegression()
model.fit(X_train, y_train)
```

▼ LinearRegression ⓘ ?

```
LinearRegression()
```

```
y_pred = model.predict(X_test)
```

```
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
import numpy as np

print("MAE:" mean_absolute_error(y_test, y_pred))
```

```
print("MSE:", mean_squared_error(y_test, y_pred))  
print("RMSE:", np.sqrt(mean_squared_error(y_test, y_pred)))  
print("R2 Score:", r2_score(y_test, y_pred))
```

MAE: 1.4607567168117603
MSE: 3.1740973539761033
RMSE: 1.78159966153345
R2 Score: 0.899438024100912

```
plt.figure(figsize=(7,5))  
plt.scatter(y_test, y_pred)  
plt.xlabel("Actual Sales")  
plt.ylabel("Predicted Sales")  
plt.title("Actual vs Predicted Sales")  
plt.show()
```

